

Enterprise Architectural Details of FashionMart

1. Data sources

- **Retail store data:** Point-of-sale (POS) systems from physical stores, which capture transaction data, inventory levels, customer purchases, and returns
- **Online web-store data:** Data from FashionMart's online store, including website traffic, customer activity, order history, product views, and abandoned carts
- **Customer data:** Data from loyalty programs, customer profiles, and behavior both in-store and online (e.g., demographics, preferences, purchase history)
- **Supply chain and inventory data:** Data from suppliers, warehouses, and logistics, including stock levels, shipment tracking, and order status
- **Marketing campaigns data:** Data from various marketing platforms (social media, email campaigns, online ads) to track engagement, conversions, and customer interactions

2. Data storage and databases

- **Customer relationship management (CRM) system:** A database to store detailed customer profiles, loyalty program data, and past interactions, helping in personalization efforts
- **Show room sales database:** Storage system for managing product data of POS systems including sales, descriptions, prices, sizes, and stock availability
- **Data warehouse1:** A centralized repository that stores structured data from multiple sources (POS, CRM, inventory, etc.). This enables efficient reporting, analytics, and data aggregation.
- **Website sales database:** Storage system for managing website data including sales, product descriptions, prices, images, sizes, and stock availability
- **Data warehouse2:** A repository that stores structured data from website. This enables to analyze online customer behavior.

3. Data integration layers

- **Data ingestion layer:** This layer collects raw data from various sources (physical stores, online, supply chain, etc.) and processes it into a usable format.
- **Data transformation layer:** This layer cleans, normalizes, and structures raw data so it can be analyzed and used effectively across the organization. ETL (extract, transform, load) tools are used for this purpose.
 - **Example:** Transforming POS data into a common format that aligns with online store transaction data, ensuring consistency across all systems.

4. Analytics and reporting layer

- **Business intelligence (BI) tools:** Dashboards and reporting tools for managers to track KPIs such as sales performance, customer engagement, inventory levels, and marketing ROI
 - **Example:** Tools such as Power BI, Tableau, or Looker can be used to create visual reports and analyze performance.

5. Inventory and supply chain management

- **Inventory management system (IMS):** A central system that tracks product availability, order status, stock levels, and supplier orders
 - **Example:** Integration with suppliers and warehouses to maintain up-to-date stock availability across all stores and online channels
- **Warehouse management system (WMS):** Ensures that the inventory data is synchronized between online orders, physical stores, and warehouses, helping with order picking, packing, and shipping

6. Security and privacy layer

- **Access control:** Role-based access control (RBAC) ensures that only authorized personnel can access sensitive data, such as customer profiles or inventory data

Summary of FashionMart's data architecture components:

1. **Data sources:** Retail Store Data, Web Site Data, Customer Data, Supply Chain Data, Marketing Campaign Data
2. **Data storage:** CRM, PIM, Data Warehouse, Cloud Storage, Real-Time Data Streaming
3. **Data integration:** Data Ingestion, Transformation, API Gateway
4. **Analytics:** BI Tools, Predictive Analytics Engine, Real-Time Analytics
5. **Inventory & supply chain:** IMS, OMS, WMS
6. **Security & privacy:** Data Encryption, Access Control, Compliance